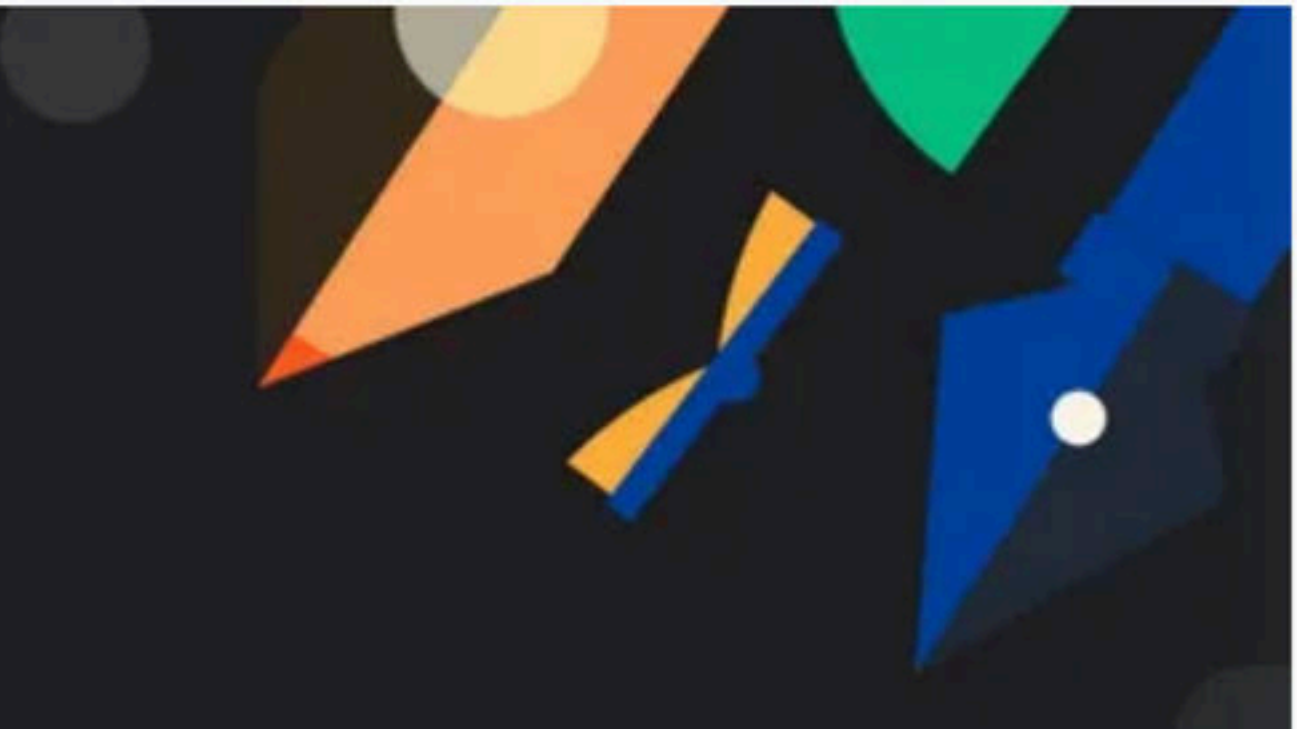
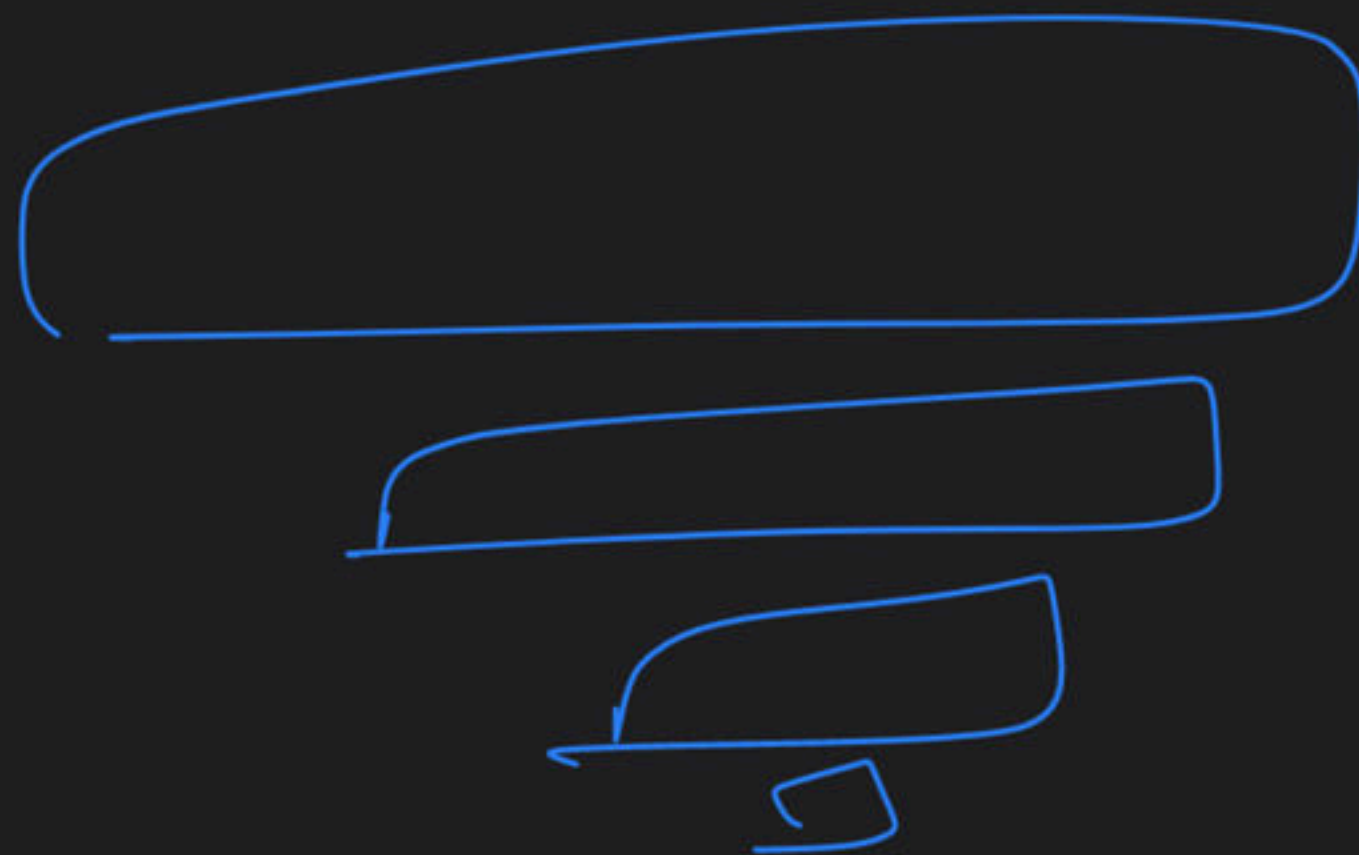
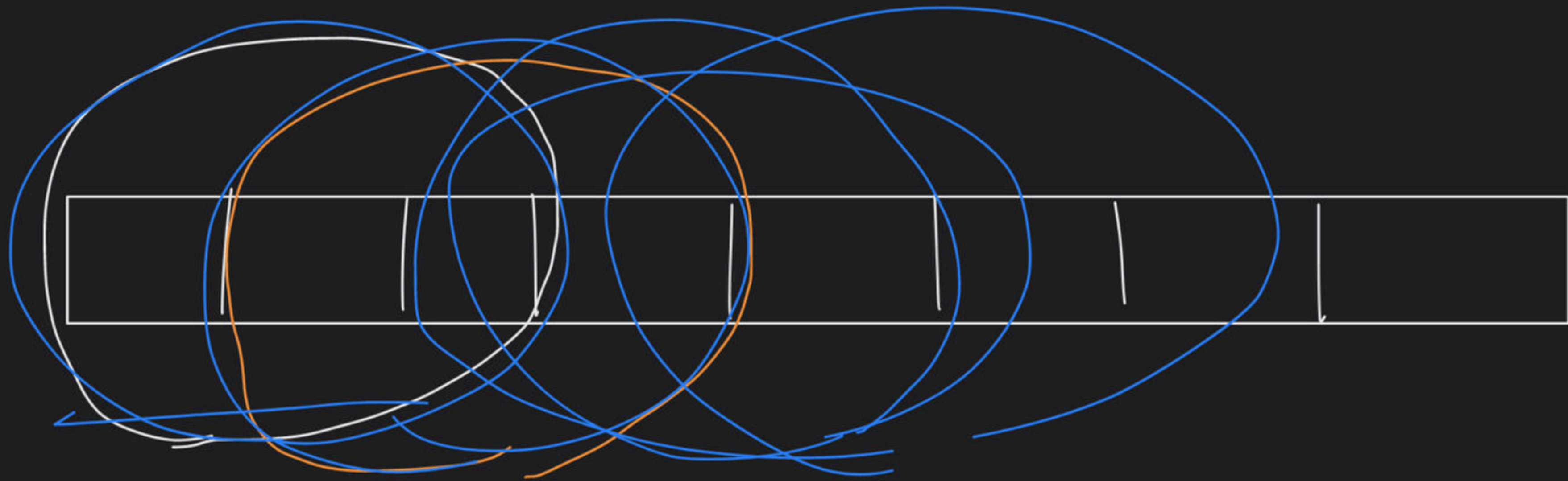




Sliding Window BONUS Session

Special class



① Sliding Window

Types



fixed-sized window

optimisation

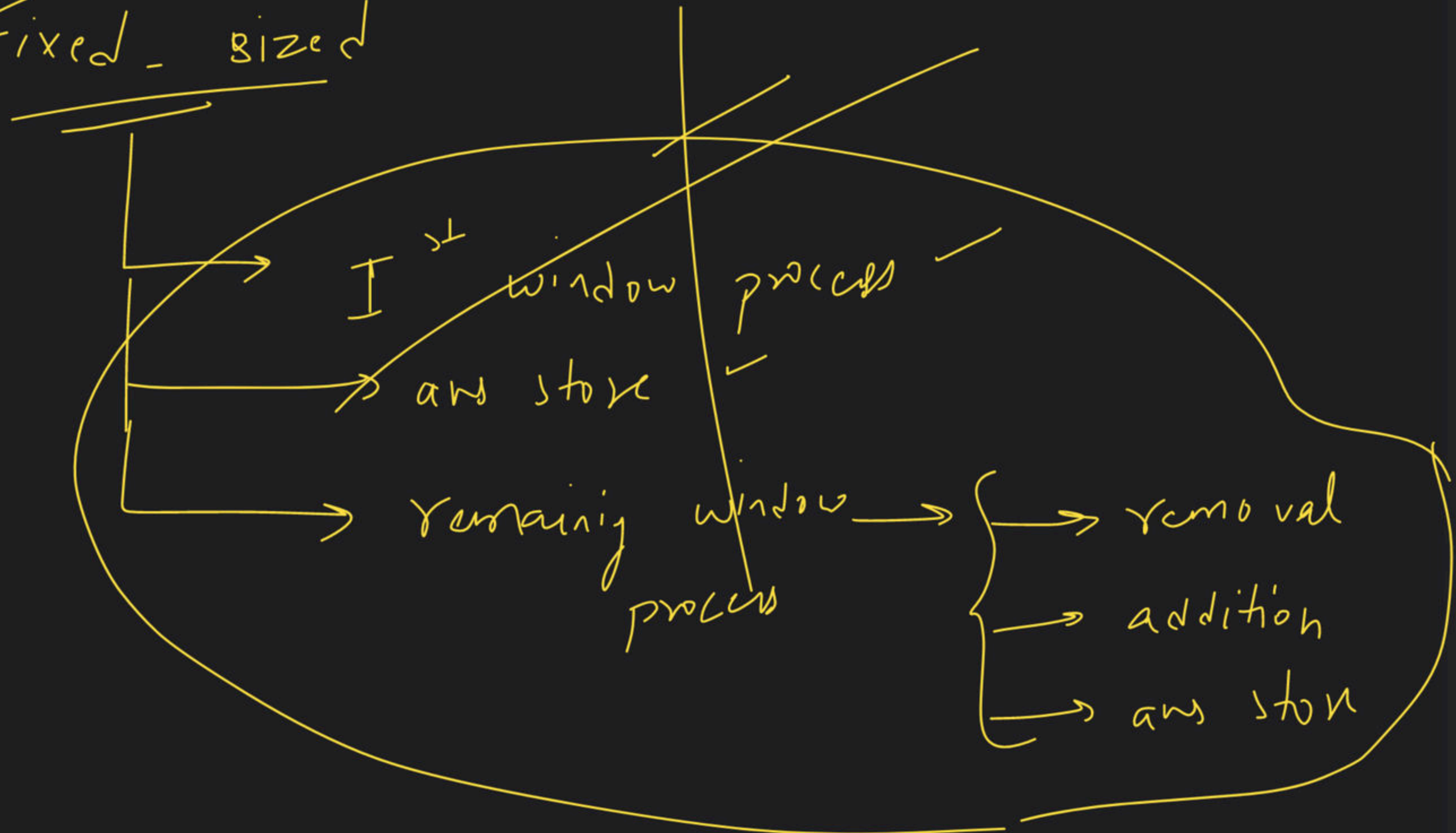


variable-size window

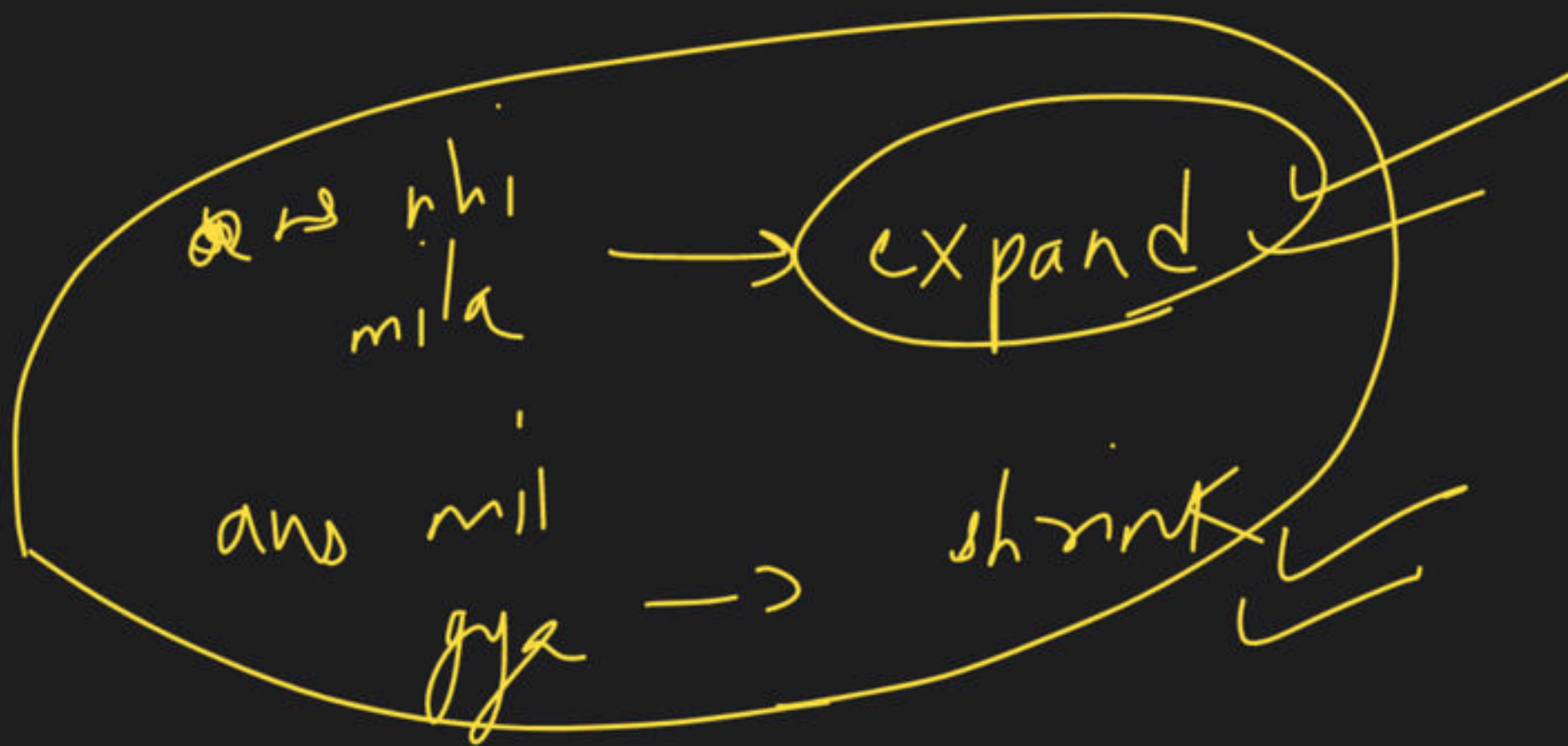
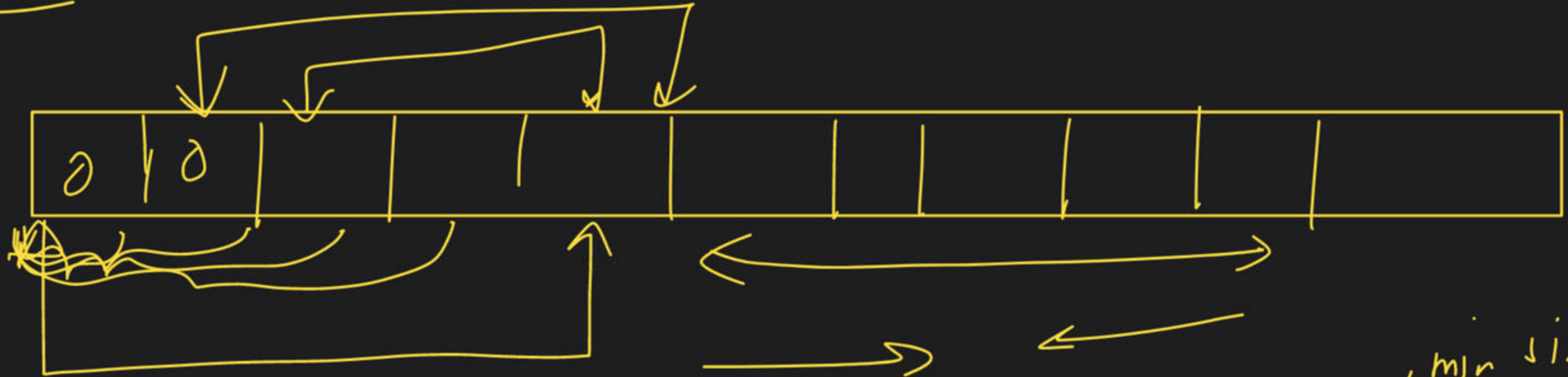


2 ptr approach

→ Fixed - sized

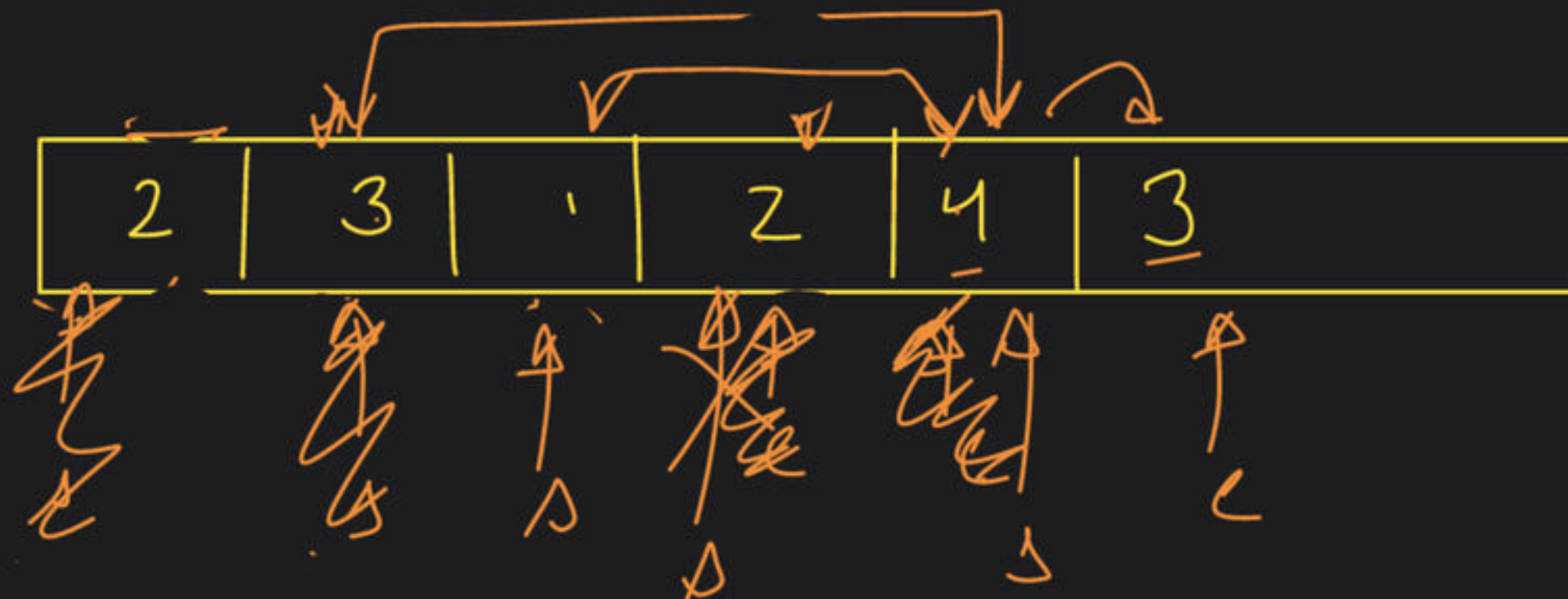


→ Variable - size window



subarray → min size window
→ Target

target = 7

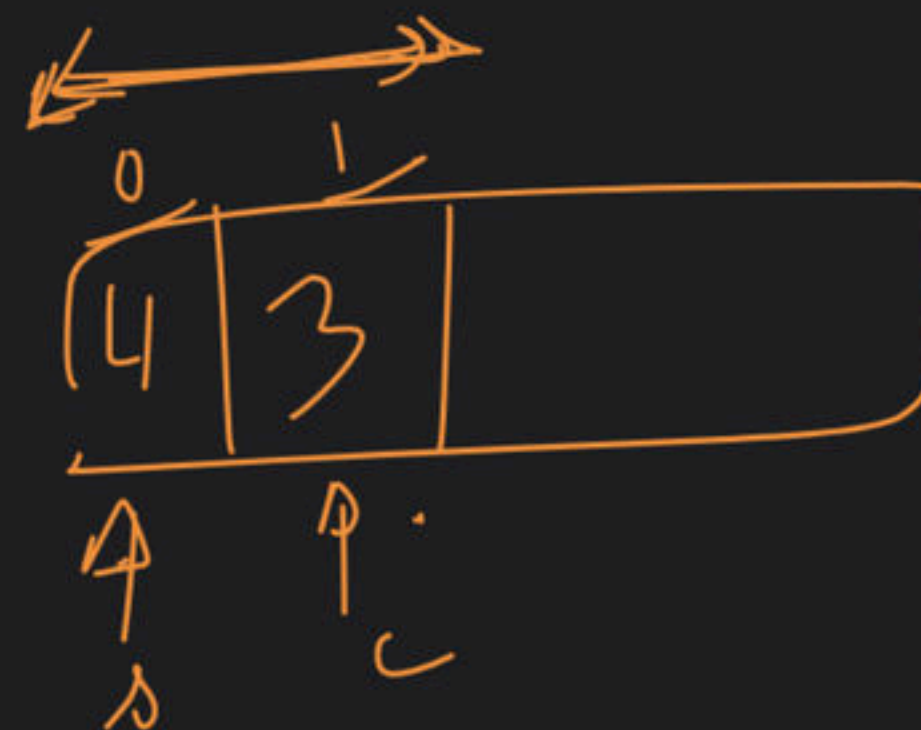


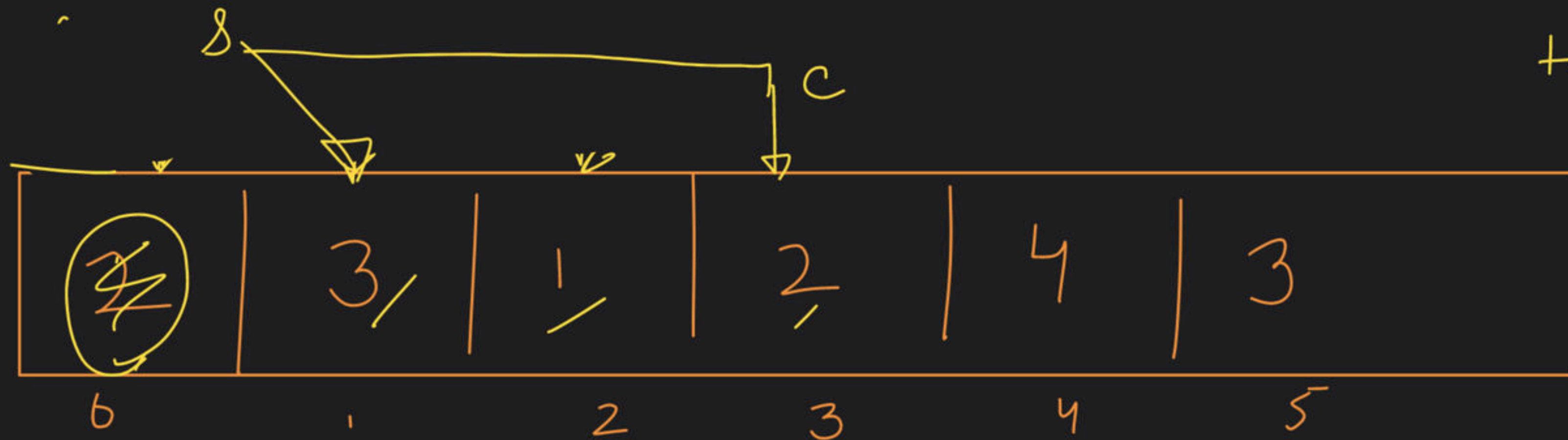
sum < target → expand → e++

sum ≥ target → shrink → s++
↓
ans store

2 - 1 + 1 = 2
e - s + 1 = 1 + 1
1 - 0 = 1

~~4~~
~~3~~
2



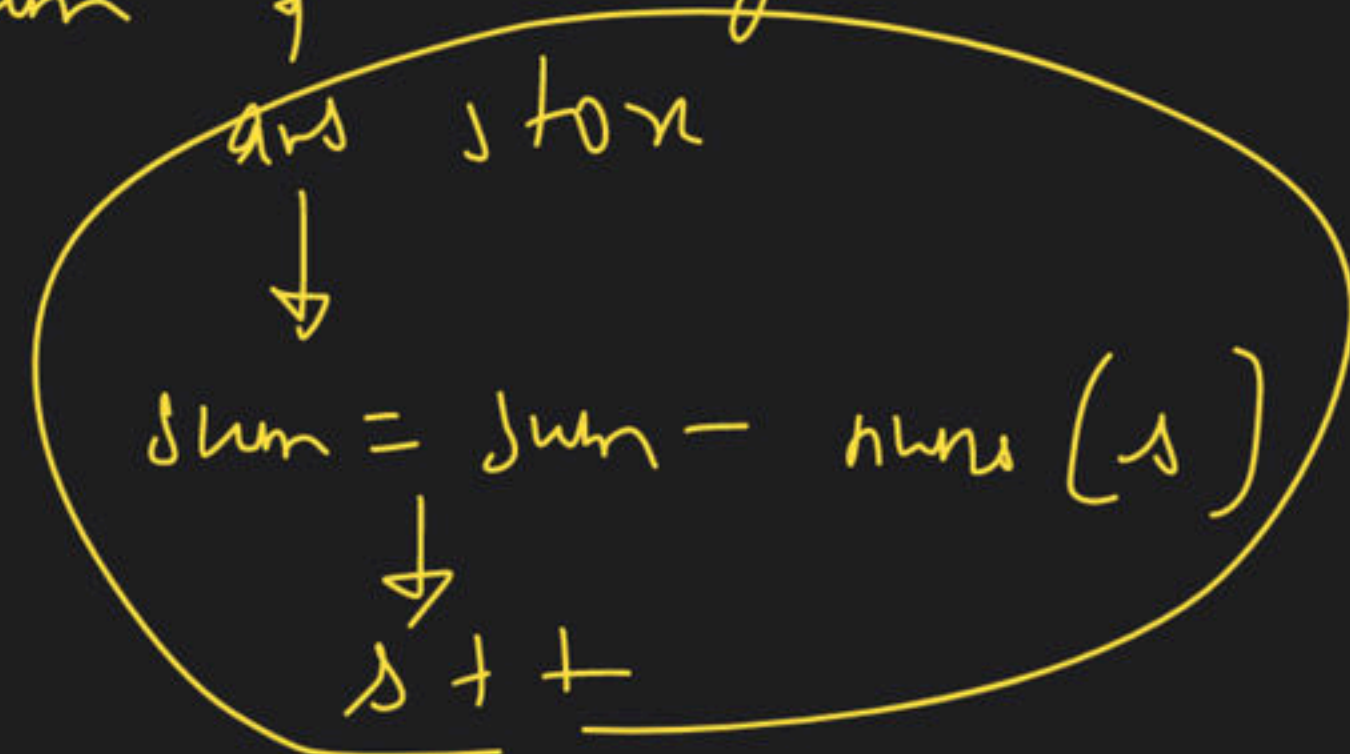


$$\text{sum} = 0 + 2 = 2 + 3 = 5 + 1 = 6 + 2 = 8 > 7$$

Since $8 > 7$, the element at index 0 (2) is removed from the window.

$$\text{sum} = 8 - 2 = 6$$

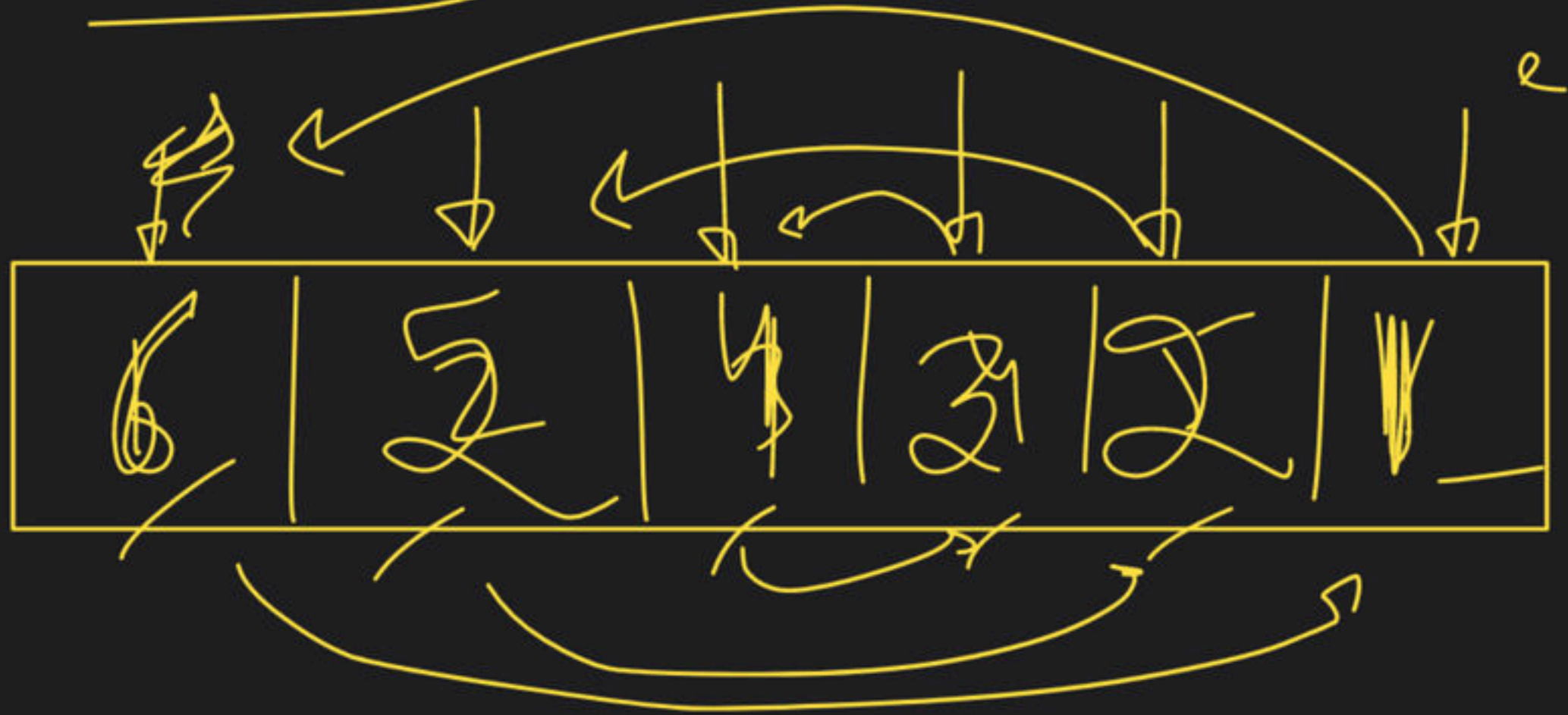
start++



len
↓
4

→ 2 pointer approach

approach :-



2 sum

2	7	11	15
0	1	2	3

#

(2, 0)	(7, 1)	(11, 2)	(15, 3)
--------	--------	---------	---------

sort

s

e

$$2 + 15 \rightarrow 17 \rightarrow \times$$

$$\underline{17} > 9 \rightarrow \dots$$

$$\text{target} = 9$$

$$2 + 7 \rightarrow 9 \checkmark$$

(0, 1)

$$2 + 11 = 13 \rightarrow 9 \times$$

$$13 > 9 \rightarrow \dots$$

mem

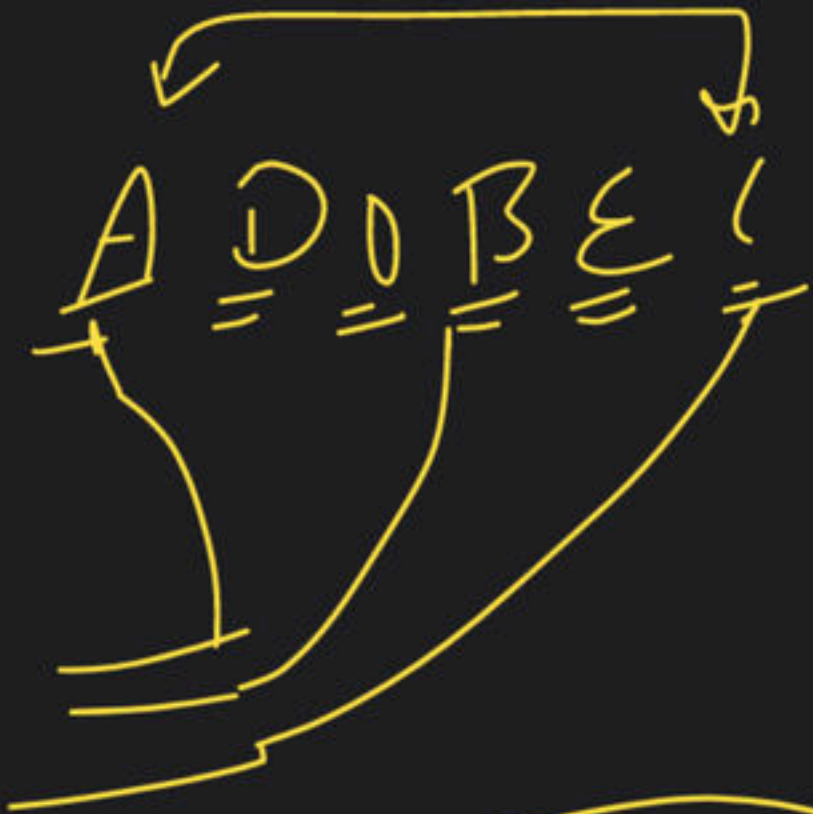
s



e



~



expand

matching characty

< h

Shrink → all store

acc. up date -> s++

A B O B C C
C O D E B A
B A N C

52

Bit Manipulation

















